

# Weighted Support Criteria for Reciprocal Mersenne Subseries

The series  $\sum_{a \in A} (b^a - 1)^{-1}$  and Erdős Problem #257

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## ABSTRACT

For a finite nonempty set  $P$  of primes, let  $h(a)$  be the  $P$ -part of  $a$ . We prove that  $\sum_{a \in A} h(a) / [a(2^{h(a)} - 1)] < \infty$  makes  $\sum_{a \in B} (b^a - 1)^{-1}$  irrational for every integer  $b \geq 2$  and every infinite  $B \subseteq A$ . The condition permits a divergent reciprocal sum, but excludes full support and all odd exponents. The proof averages positive displacements over multiples of a finite modulus; a second, dyadic average controls incomplete periods. We also prove the reciprocal-summable criterion stated by Erdős and a common-average extension using positive divisor covers. The two sufficient classes are incomparable; their mixed criterion admits supports in neither class. Supplementary results concern finite denominators and conditional rational-membership tests, not a resolution of the universal problem.

## 1 Introduction and main results

For  $A \subseteq \mathbb{N}_{>0}$  and  $b > 1$ , put  $X_A(b) = \sum_{a \in A} (b^a - 1)^{-1}$ . This converges because  $(b^a - 1)^{-1} \leq b^{1-a} / (b - 1)$ . For a finite set  $P$  of primes, the  $P$ -part of  $a$  is  $h(a) = \prod_{p \in P} p^{v_p(a)}$ , where  $v_p(a)$  is the exponent of  $p$  in  $a$ . For example, if  $P = \{2\}$  and  $a = 2^k m$  with  $m$  odd, then  $h(a) = 2^k$ .

**Theorem 1.1** (a weighted condition on the support). *Let  $b \geq 2$  be an integer, let  $A \subseteq \mathbb{N}_{>0}$  be infinite, and let  $P$  be a finite nonempty set of primes. Set  $h(a) = \prod_{p \in P} p^{v_p(a)}$ . If*

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$$W_{b,P}(A) := \sum_{a \in A} \frac{h(a)}{a(b^{h(a)} - 1)} < \infty, \quad (1)$$

*then  $X_A(b)$  is irrational. In particular, the base-two condition*

$$\sum_{a \in A} \frac{h(a)}{a(2^{h(a)} - 1)} < \infty \quad (W)$$

*implies that  $X_A(b)$  is irrational for every integer  $b \geq 2$ . Both conclusions are hereditary under passage to infinite subsets.*

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\**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

For  $P = \{2\}$ , the summand in (1) is  $1/[m(b^{2^k} - 1)]$  when  $a = 2^k m$  and  $m$  is odd. Large powers of 2 can therefore compensate for large reciprocal mass among the odd factors. The example following the proof in Section 3 makes this precise. The reciprocal mass at each fixed  $P$ -part must be finite, and its weighted sum over the parts must converge; the first requirement alone is insufficient. Section 1.2 of the companion, *Reciprocal Mersenne Subseries*, gives the decomposition and a separating example.

For every finite  $P$ , full support and all odd exponents fail the condition: the integers coprime to  $2 \prod_{p \in P} p$  have  $h(a) = 1$  and divergent reciprocal sum, by inclusion–exclusion. The full set of primes fails too, since the primes outside  $P$  contribute  $1/[a(b - 1)]$ . These are limitations of this criterion, not assertions of rationality. By contrast,  $h/(2^h - 1) \leq 1$  shows that it includes every reciprocal-summable support.

**Theorem 1.2** (reciprocal-summable supports). *Let  $A \subseteq \mathbb{N}_{>0}$  be infinite. If*

$$\sum_{a \in A} \frac{1}{a} < \infty,$$

*then  $X_A(b)$  is irrational for every integer  $b \geq 2$ .*

Erdős stated Theorem 1.2, including the all-base conclusion, after proving its pairwise-coprime case [7, p. 222]. We give a complete averaging proof of that stated extension. Erdős also discussed weakening reciprocal summability [7, pp. 222, 226]; no identification of the weighted condition with his suggested conditions is asserted.

The proof uses a positive displacement that rationality would separate from zero. Erdős likewise points to fractional parts of powers times the series value [7, p. 226]. For an integer  $b \geq 2$ , infinite  $A$  and  $N > 0$ , division of  $N$  by each  $a \in A$  gives

$$0 < \Delta_{b,A}(N) := \sum_{a \in A} \frac{b^{N \bmod a} - 1}{b^a - 1} = (b^N - 1)X_A(b) - J_{b,A}(N), \quad J_{b,A}(N) \in \mathbb{Z}. \quad (\text{D})$$

The integer term is explicitly

$$J_{b,A}(N) = \sum_{\substack{a \in A \\ a \leq N}} \sum_{j=1}^{\lfloor N/a \rfloor} b^{N-ja}.$$

The identity holds for every real  $b > 1$ ; integrality uses the integer-base hypothesis. Every summand in the displacement is nonnegative, and an exponent  $a > N$  gives a positive summand; such an exponent exists because  $A$  is infinite. Rationality  $X_A(b) = p/q$  would force every positive displacement to be at least  $1/q$ . The proof makes these positive displacements arbitrarily small by averaging along multiples of a growing divisibility modulus.

For an application of Theorem 1.2, call an integer *powerful* when every prime dividing it occurs to exponent at least two. Every such integer can be written as  $u^2 v^3$ , so

$$\sum_{\substack{a \geq 1 \\ a \text{ powerful}}} \frac{1}{a} \leq \zeta(2)\zeta(3) < \infty.$$

Thus arbitrary infinite thinnings of the powerful integers are included. For full perfect-power supports, compare Duverney–Tachiya [1, Corollary 1.2, p. 4] and the earlier linear-independence results of Luca–Tachiya [13].

**Problem 1.3** (Erdős #257). Is  $X_A(2)$  irrational for every infinite  $A \subseteq \mathbb{N}_{>0}$ ?

Sections 2–3 prove the support criteria: first the simpler reciprocal-summable argument, then the weighted theorem and its common-average extension. The remaining sections give supplementary finite-denominator and rational-membership results. None is needed for the weighted proof.

*Formal proofs.* A result with a kernel-checked Lean proof carries a mark in the margin. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. *Comparator* opens the record of an independent check, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean’s Comparator tool, allowing only the three standard axioms. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text.

The coefficient sequence throughout is the divisor transform of the indicator function of the support:

$$c_A(n) = \#\{a \in A : a \mid n\}, \quad X_A(b) = \sum_{n \geq 1} \frac{c_A(n)}{b^n}. \quad (2)$$

Tonelli’s theorem applied to  $(b^a - 1)^{-1} = \sum_{j \geq 1} b^{-aj}$  proves the identity. Moreover  $0 \leq c_A(n) \leq \tau(n)$ . The selector  $1_A$  is Boolean; its divisor transform  $c_A$  generally is not.

## 2 Reciprocal-summable supports at every integer base

A shift divisible by each exponent in a finite part of  $A$  makes those terms of the displacement vanish. We then average over such shifts. Reciprocal summability controls the terms outside that finite part without requiring a common period for the whole support.

Fix the integer base  $b \geq 2$  and put

$$w_{b,d}(N) = \frac{b^{N \bmod d}}{b^d - 1}, \quad T_N^{(b)} = \sum_{d \in A} w_{b,d}(N).$$

Then  $T_N^{(b)} - T_0^{(b)} = \Delta_{b,A}(N)$ . For a fixed positive integer  $Q$ , the orbit of  $Q$  modulo  $d$  consists of  $d/g$  residues, where  $g = (Q, d)$ . Summing the geometric progression gives

$$M_{Q,b}(d) := \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{m=1}^X w_{b,d}(Qm) = \frac{g}{d(b^g - 1)} \leq \frac{1}{d}. \quad (3)$$

For example, at  $b = 2$ ,  $Q = 4$  and  $d = 6$ , the orbit is 4, 2, 0 modulo 6. Its mean atom is  $(16 + 4 + 1)/(3 \cdot 63) = 1/9$ , whereas its zero-shift atom is  $1/63$ . If  $6 \mid Q$ , the orbit instead stays at zero and the mean is exactly  $1/63$ . Making an exponent divide  $Q$  therefore changes its orbit mean. The infinite tail still requires the bound that follows. The passage from individual atoms to their infinite sum needs a uniform bound. Counting positive multiples of  $d$  gives

$$\begin{aligned} \frac{1}{X} \sum_{m=1}^X w_{b,d}(Qm) &= \sum_{r \geq 1} b^{-r} \frac{\#\{1 \leq m \leq X : d \mid Qm + r\}}{X} \\ &\leq \sum_{r \geq 1} b^{-r} \frac{Q+r}{d} = \frac{Q/(b-1) + b/(b-1)^2}{d} \leq \frac{Q+2}{d}. \end{aligned}$$

Indeed, the counted multiples are distinct and at most  $QX + r$ . For this fixed  $Q$ , the majorant  $(Q+2)/d$  is summable over  $A$ . Dominated convergence therefore takes the observation-length limit inside the sum over  $d$ , giving the Cesàro mean  $\sum_{d \in A} M_{Q,b}(d)$ . This majorant is not uniform in growing  $Q$ .

Next let  $Q_t = \text{lcm}(1, \dots, t)$ . For each fixed  $d$ , eventually  $d \mid Q_t$  and  $M_{Q_t,b}(d) = (b^d - 1)^{-1}$ . A second dominated-convergence argument, using  $M_{Q_t,b}(d) \leq 1/d$ , yields

$$\sum_{d \in A} M_{Q_t,b}(d) \rightarrow X_A(b). \quad (4)$$

Thus the limiting averages of the nonnegative displacements  $\Delta_{b,A}(Q_t m)$  tend to zero. Choose  $t$ , then a sufficiently long finite average, and finally a term no larger than that average. This gives arbitrarily small positive displacements, contradicting the rational lattice in (D). This proves Theorem 1.2 directly at every integer base.

The order of limits is essential: the observation length tends to infinity with the modulus fixed, and only then does the modulus increase. Reciprocal summability controls both interchanges.

### 3 Extensions beyond reciprocal summability

The weighted condition can hold when  $\sum_{a \in A} 1/a$  diverges, so the previous proof's summable majorant is no longer available. We instead bound finite averages for each exponent and then average over dyadic observation lengths. This second average controls the incomplete periods. Estimate (S) will also allow the divisor-cover argument to use these same observation lengths.

#### 3.1 Proof of the weighted criterion

For a finite nonempty prime set  $P$ , put  $h(a) = \prod_{p \in P} p^{v_p(a)}$ .

We prove the fixed-base assertion of Theorem 1.1. The base-two condition implies every integer-base instance because  $b^{h(a)} - 1 \geq 2^{h(a)} - 1$ .

*Proof.* The rational-lattice obstruction is (D): if  $X_A(b) = p/q$ , every positive  $\Delta_{b,A}(m)$  is at least  $1/q$ . For  $d_a(m) = (b^{m \bmod a} - 1)/(b^a - 1)$  and  $g = (Q, a)$ , one complete orbit of

$Qm \bmod a$  gives, for  $Q, a, T \geq 1$ ,

$$\frac{1}{T} \sum_{t=1}^T d_a(tQ) \leq \frac{g}{a(b^g - 1)} + \frac{1}{T(b^g - 1)}. \quad (5)$$

Indeed, the orbit has length  $a/g$  and  $\sum_{t=1}^{a/g} b^{tQ \bmod a} / (b^a - 1) = 1/(b^g - 1)$ ; an incomplete orbit costs at most one more complete orbit. Also, for  $Y = QT$ ,

$$\frac{1}{T} \sum_{t=1}^T \sum_{\substack{a \in A \\ a > Y}} d_a(tQ) \leq \frac{4}{T}, \quad (6)$$

because  $d_a(tQ) \leq 2 \cdot 2^{tQ-a}$  when  $a > QT$  and  $\sum_{t=1}^T 2^{tQ-QT} \leq 2$ .

The incomplete-orbit errors are controlled by a second finite average. For integers  $Q, M \geq 1$  and  $\alpha_a \geq 0$  with  $\sum_a \alpha_a / a < \infty$ ,

$$\sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{a \leq Q2^j} \alpha_a \leq 2Q \sum_{a \geq 1} \frac{\alpha_a}{a}; \quad (7)$$

for each fixed  $a$ , the admissible geometric tail is at most  $2Q/a$ .

Fix  $\varepsilon > 0$ . Choose finite nonempty  $F \subseteq A$  so that the (1) mass outside  $F$  is below  $\varepsilon$ , and let  $L$  be any fixed positive common multiple of  $F$ . With  $p_* = \max P$ ,  $r = |P|$ , and large  $H \geq 2p_*$ , set

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \quad G = \left\lfloor \frac{H}{p_*} \right\rfloor.$$

For  $a \in F$  the displacement term vanishes, since  $a \mid Q$ . For  $a \notin F$  with  $h(a) \leq H$ , we have  $h(a) \mid Q$  and hence  $(Q, a) \geq h(a)$ . The complete-period term in (5) is therefore bounded by  $h(a)/[a(b^{h(a)} - 1)]$ , using the monotonicity of  $n/(b^n - 1)$ . Their sum is less than  $\varepsilon$ ; their incomplete-period errors will be handled by the dyadic average.

For  $h(a) > H$ , we instead have  $(Q, a) \geq G$ . Sum the complete-period bounds over  $a \leq QT$ , using  $\sum_{a \leq QT} 1/a \leq 1 + \log(QT)$ . There are at most  $QT$  incomplete-period terms, each at most  $1/[T(b^G - 1)]$ . The total contribution from these exponents is thus at most

$$\frac{G(1 + \log(QT)) + Q}{b^G - 1}.$$

For the gcd claim, either every  $P$ -prime-power component of  $h(a)$  is at most  $H$ , in which case  $h(a) \mid Q$ , or some  $p^{v_p(a)} > H$  contributes  $p^{\lfloor \log_p H \rfloor} > H/p \geq H/p_* \geq G$  to the gcd. The second averaging length must make both  $Q/M$  and  $GM/b^G$  small. Since  $Q$  grows only polynomially in  $H$  and  $G$  grows linearly,  $M = \lfloor b^{G/2} \rfloor$  lies between these two scales. Combining the preceding estimates with (6), averaging over  $T = 2^j$  for  $M \leq j < 2M$ , and using (7) with  $\alpha_a = \mathbf{1}_A(a)/(b^{h(a)} - 1)$  yields

$$\frac{1}{M} \sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{t=1}^{2^j} \Delta_{b,A}(tQ) \leq \varepsilon + \frac{2QW_{b,P}(A)}{M} + \frac{G(1 + \log Q + 2M \log 2) + Q}{b^G - 1} + 4 \cdot 2^{-M}. \quad (8)$$

The order of choices is important. First fix  $\varepsilon$  and choose  $F$  and its common multiple  $L$ ; only then let  $H \rightarrow \infty$ . With  $L$  fixed,  $Q \leq LH^r$ ,  $G = H/p_* + O(1)$  and  $M \asymp b^{G/2}$ , so every term after  $\varepsilon$  tends to zero. This produces arbitrarily small displacements, but gives no specified decay rate in the index  $N = Qt$ . The left side is a finite average of positive displacements, so one is below  $2\varepsilon$  for large  $H$ . Since  $\varepsilon$  is arbitrary, this contradicts the lower bound  $1/q$  under rationality. Finally  $W_{b,P}(A) \leq W_{2,P}(A)$  for  $b \geq 2$ , and weighted mass decreases on taking subsets.  $\square$

### 3.2 An example beyond reciprocal summability

The hypothesis is genuinely weaker than reciprocal summability. Let

$$A_* = \{2^k m : k \geq 1, m \text{ odd}, m \leq 2^{2^k}\}.$$

The layers are disjoint, since their elements have different 2-adic valuations. For  $r \geq 2$ , put

$$S_r = \sum_{\substack{m \leq 2^r \\ m \text{ odd}}} \frac{1}{m}.$$

Each block  $2^j \leq m < 2^{j+1}$ ,  $1 \leq j < r$ , contains  $2^{j-1}$  odd integers, with reciprocal sum between  $1/4$  and  $1/2$ . Including  $m = 1$  gives  $r/4 \leq S_r \leq r$ . The ordinary reciprocal mass of the  $k$ th layer is  $2^{-k} S_{2^k} \geq 1/4$ , so  $\sum_{a \in A_*} 1/a$  diverges. For  $P = \{2\}$ , that layer's weighted mass satisfies

$$\frac{S_{2^k}}{2^{2^k} - 1} \leq \frac{2^k}{2^{2^k} - 1} \leq 2^{1-k}.$$

The last bound uses  $2^{2^k} - 1 \geq 2^{2^k - 1}$  and  $2^k \geq 2k$ . The weighted mass is summable. Every infinite subset of  $A_*$  therefore has irrational  $X_A(b)$  at every integer base, although  $\sum_{a \in A_*} 1/a$  diverges. This example is an ordinary argument; the Lean proofs do not treat this particular set.

Both proofs select a term no larger than a finite average. For a related selection step on arithmetic progressions, see Duverney–Tachiya [1, Section 2, (2.3)–(2.9)]. The sparse-coefficient criteria of Kaneko–Suzuki–Tachiya [2, Theorems 1 and 3] do not apply directly to  $c_A$ : for nonempty  $A$ , it is positive on every multiple of  $\min A$ . The companion, Section 1.2, gives the counting argument and distinguishes their remote-tail average from the displacement used here.

One host realises any monotone rule on a finite set  $E$  of primes: if  $\mathcal{U}$  is an upward-closed family of subsets of  $E$  with  $E \in \mathcal{U} \not\supset \emptyset$ , a finite union of prime-cofactor blocks has divergent reciprocal sum and  $W_{b,P} < \infty$  exactly when  $P \cap E \in \mathcal{U}$ , for every  $b \geq 2$  and finite prime set  $P$ , so every infinite subset is irrational at every base. The companion proves this and Lean checks it.

### 3.3 A common finite average for the extensions

To combine the weighted criterion with positive divisor majorants, both arguments must use the same indices. The following estimate supplies that common average; Theorem 3.3 gives the combination.

For  $1 < B \leq 2$ , positive integers  $L, d, M$ , and an integer  $R \geq 0$ , put

$$w_{B,d}(n) = \frac{B^{n \bmod d}}{B^d - 1}, \quad \mathcal{D}_{L;R,M} F = \frac{1}{M} \sum_{j=R}^{R+M-1} \frac{1}{2^j} \sum_{m=1}^{2^j} F(Lm).$$

The finite estimate

$$\mathcal{D}_{L;R,M} w_{B,d} \leq \frac{1 + 4L/M}{d(B-1)} \quad (\text{S})$$

equivalently bounds  $(B-1)\mathcal{D}_{L;R,M} w_{B,d}$  by  $(1 + 4L/M)/d$ . This normalized bound is uniform as  $B \downarrow 1$ ; the unnormalized right side grows like  $(B-1)^{-1}$ . The ratio  $L/M$  measures the cost of incomplete modular periods.

To prove it, fix  $T = 2^j$  and average over  $1 \leq m \leq T$ . When  $d \leq LT$ , the complete orbit has length  $d/g$ , with  $g = (L, d)$ , and total weight  $1/(B^g - 1)$ . Complete cycles and one remaining piece give

$$\frac{1}{T} \sum_{m=1}^T w_{B,d}(Lm) \leq \frac{g}{d(B^g - 1)} + \frac{1}{T(B^g - 1)} \leq \frac{1}{d(B-1)} + \frac{1}{T(B-1)}.$$

Across dyadic lengths satisfying  $d \leq L2^j$ , the reciprocal-length errors sum to at most  $2L/[d(B-1)]$ . When  $d > 2LT$ , there is no wrap and  $2Lm \leq d-1$ ; hence  $\sum_{i=0}^{d-1} B^i \geq dB^{(d-1)/2} \geq dB^{Lm}$ . Each atom is then at most  $1/[d(B-1)]$ . Finally, at most one dyadic length satisfies  $LT < d \leq 2LT$ . For that length the geometric sum and convexity give

$$\frac{1}{T} \sum_{m=1}^T w_{B,d}(Lm) = \frac{B^L}{T(B^L - 1)} \frac{B^{LT} - 1}{B^d - 1} \leq \frac{LB^L}{d(B^L - 1)} \leq \frac{2L}{d(B-1)}.$$

For the last inequality, use  $(B^L - 1)/(B - 1) = \sum_{i=0}^{L-1} B^i \geq B^{L-1}$  and  $B \leq 2$ . Summing the main terms and the two error bounds proves (S). This is an ordinary proof; the estimate is also [proved in Lean](#). Nonnegative interchange permits summation against any coefficients  $c_d$  with  $\sum_d c_d/d < \infty$ .

### 3.4 Positive divisor majorants

For a finite set  $F$ , write  $f_F(n) = \#\{a \in F : a \mid n\}$ . We bound a fractional power of this count by a nonnegative sum over divisors of  $n$ . The inequality must hold for every positive integer  $n$ , not just on average.

**Theorem 3.1** (a summable divisor-cover criterion). *For each  $j \geq 1$ , let  $F_j \subseteq \mathbb{N}_{>0}$  be finite, let  $0 < \alpha_j \leq 1$ , and let  $c_{j,d} \geq 0$  satisfy*

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$$f_{F_j}(n)^{\alpha_j} \leq \sum_{d \mid n} c_{j,d} \quad (n \geq 1).$$

Set  $C_j = \sum_{d \geq 1} c_{j,d}/d$ . If

$$\sum_{j \geq 1} \frac{C_j 2^{j\alpha_j}}{2^{\alpha_j} - 1} < \infty, \quad (\text{V})$$

then  $X_A(b)$  is irrational for every infinite  $A \subseteq \bigcup_j F_j$  and every integer  $b \geq 2$ .

Every finite set admits a majorant of the required kind: take  $\alpha_j = 1$  and  $c_{j,d} = \mathbf{1}_{F_j}(d)$ . The restriction is that the costs of the whole sequence satisfy (V). For a simple admissible family, take  $F_j = \{4^j\}$ ,  $\alpha_j = 1$ , and  $c_{j,4^j} = 1$ , with all other coefficients zero. Then  $C_j = 4^{-j}$  and the series in (V) is  $\sum_j 2^{-j}$ . The logarithmic obstruction below gives a necessary condition for such a cover.

*Proof.* Write  $B_j = 2^{\alpha_j}$  and

$$U_j(N) = \sum_{r \geq 1} 2^{-r} f_{F_j}(N+r), \quad V_j(N) = \sum_{d \geq 1} c_{j,d} w_{B_j,d}(N).$$

The choice  $B_j = 2^{\alpha_j}$  matches the decay after taking a fractional power. Subadditivity and the divisor majorant give

$$U_j(N)^{\alpha_j} \leq \sum_{r \geq 1} B_j^{-r} f_{F_j}(N+r)^{\alpha_j} \leq \sum_{d \geq 1} c_{j,d} \sum_{\substack{r \geq 1 \\ d|N+r}} B_j^{-r} = V_j(N).$$

The last identity sums a geometric progression in each residue class. The geometric-series identity also gives

$$\Delta_{2,F_j}(N) = U_j(N) - X_{F_j}(2) \leq U_j(N).$$

Consequently, once the first  $J$  finite sets have zero displacement, nonnegativity bounds the displacement of their union by  $\sum_{j>J} U_j(N)$ .

Fix  $\varepsilon > 0$ , set  $t_j = \varepsilon 2^{-j}$ , and choose  $J$  with

$$K_J := \sum_{j>J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{4}.$$

This is possible since  $\varepsilon^{-\alpha_j} \leq \max(1, \varepsilon^{-1})$ . Choose  $L$  divisible by every member of the first  $J$  finite sets. Equation (S) bounds the finite mean of  $S_J(N) := \sum_{j>J} t_j^{-\alpha_j} V_j(N)$  by  $(1 + 4L/M)K_J < 1/2$  whenever  $M \geq 4L$ . One sample therefore has  $S_J(N) < 1$ , forcing  $U_j(N) < t_j$  for every  $j > J$ . Every exponent in the first  $J$  finite sets divides  $L$ , so those terms have zero displacement. Consequently,

$$0 < \Delta_{2,A}(N) \leq \sum_{j>J} U_j(N) \leq \varepsilon.$$

The fixed rational lattice excludes rationality at base two. For  $0 \leq r < d$ , the function  $(b^r - 1)/(b^d - 1)$  is nonincreasing on  $b > 1$ . For  $r > 0$ , cancel  $b - 1$  and write it as  $A(b)/(A(b) + C(b))$ , where  $A(b) = \sum_{i<r} b^i$  and  $C(b) = \sum_{r \leq j < d} b^j$ . Its derivative has the sign of  $A'C - AC' = \sum_{i<r \leq j < d} (i-j)b^{i+j-1} < 0$ . Thus  $\Delta_{b,A}(N) \leq \Delta_{2,A}(N)$  for every real  $b \geq 2$ ; when  $b$  is an integer, the same rational-lattice argument applies. Any prescribed positive divisor can be included in  $L$ , so the witnesses can also be required to be arbitrarily large.  $\square$

The same proof permits any positive weights  $\eta_j$  with  $\sum_j \eta_j = 1$ : replace  $2^{j\alpha_j}$  in (V) by  $\eta_j^{-\alpha_j}$  and take  $t_j = \varepsilon \eta_j$ . The stronger condition  $\sum_j C_j 2^{j\alpha_j} 2^{\alpha_j} / (2^{\alpha_j} - 1)^2 < \infty$  implies (V), since each of its summands is the corresponding summand of (V) multiplied by  $2^{\alpha_j} / (2^{\alpha_j} - 1) > 1$ . This proves inclusion of the hypotheses for a given cover, not strict inclusion of the resulting support classes after optimisation over all covers. No separating construction is proved here.

### 3.5 What every positive cover must pay

A necessary lower bound on the cost holds for every choice of cover. For finite  $F$ , write  $\mathbb{E}_F$  for the uniform mean modulo  $\text{lcm}(F)$ , with  $\text{lcm}(\emptyset) = 1$  and  $f_\emptyset = 0$ . More generally allow weights  $\eta_j > 0$  with  $\sum_j \eta_j = 1$ , and set

$$K = \sum_j \frac{C_j \eta_j^{-\alpha_j}}{2^{\alpha_j} - 1}, \quad \Psi(0) = 0, \quad \Psi(t) = \inf_{0 < \alpha \leq 1} \frac{t^\alpha}{2^\alpha - 1} \quad (t \geq 1).$$

Here  $\log^+ t = \log \max\{1, t\}$ . For every finite  $F$  contained in  $\bigcup_j F_j$ ,

$$K \geq \mathbb{E}_F \Psi(f_F) \geq e \mathbb{E}_F \log^+ f_F. \quad (9)$$

Indeed, at a point where  $f_F(n) = t > 0$ , some covering set  $F_j$  satisfies  $f_{F_j}(n) \geq \eta_j t$ . Otherwise summing contradicts coverage. The corresponding weighted majorant is at least  $\Psi(t)$ . Average first over  $1 \leq n \leq X$ , using  $\lfloor X/d \rfloor / X \leq 1/d$ , and let  $X \rightarrow \infty$ . The periodic left side tends to  $\mathbb{E}_F \Psi(f_F)$ , proving the first inequality; the covering moduli need not divide  $\text{lcm}(F)$ . Convexity gives  $2^\alpha - 1 \leq \alpha$ , and  $e^u/u \geq e$  proves the second. Those scalar steps, including the bound  $t^\alpha / (2^\alpha - 1) \geq e \log t$ , are [proved in Lean](#); the assembled averaging argument is an ordinary proof.

This bound survives optimisation over all covers. For  $F(q, P) = \{qd : d \mid \prod_{p \in P} p\}$ , where  $q \geq 2$  and no  $p \in P$  divides  $q$ , put  $S = \sum_{p \in P} 1/p$ . If  $S \geq 1$ , the infimum  $K_*$  over finite or countable covers satisfies

$$\frac{e(S-1)}{q} \leq K_*(F(q, P)) \leq \frac{eS}{q}. \quad (10)$$

For the lower bound, condition on  $q \mid n$  and write  $f_F(n) = 2^z$ . The Chinese remainder theorem gives  $\mathbb{E}Z = S$ . For  $z \geq 0$  and  $v = \alpha \log 2 > 0$ ,

$$\frac{2^{\alpha z}}{2^\alpha - 1} \geq \frac{e^{(z-1)v}}{v} \geq e(z-1) \quad (z > 1);$$

for  $0 \leq z \leq 1$  the claimed lower bound is nonpositive. Thus  $\Psi(2^z) \geq e(z-1)$ . For the upper bound, use the one-set cover with  $z = 1/S$  and  $\alpha = \log_2(1+z)$ ; its exact positive expansion has cost

$$\frac{1}{qz} \prod_{p \in P} (1 + z/p) \leq \frac{eS}{q}.$$

In particular,  $1 - 1/S \leq qK_*(F(q, P)) / (eS) \leq 1$ . Thus  $K_*(F(q, P)) \sim eS/q$  as  $S \rightarrow \infty$ , uniformly over the permitted choices of  $q$  and  $P$ .

The logarithmic lower bound is not a converse: replacing a fractional majorant by a logarithmic one need not control averages along the multiples of a prescribed modulus. A counterexample and the distinct bound for ordinary initial intervals are proved in the companion, Section 13; those proofs were first worked out in an AI-assisted note of 17 September 2026. The three results of that section also have Lean proofs, each compared independently by Comparator; the ordinary proofs have not had independent human review. Those arguments concern finite functionals. The next construction uses the divisor cubes to compare the infinite-support criteria. Write  $\mathcal{W}_b$  for the supports satisfying the finite-prime weighted hypothesis at base  $b$ , and  $\mathcal{C}$  for supports admitting a strengthened positive cover as in (V).

**Proposition 3.2** (incomparable support criteria). *There are infinite positive supports  $E$  and  $V$  such that*

$$E \in \mathcal{W}_2, \quad E \notin \mathcal{C}, \quad V \in \mathcal{C}, \quad V \notin \mathcal{W}_b \ (b \geq 2), \quad \sum_{a \in V} \frac{1}{a} = \infty.$$

*Proof.* The reciprocal sum over odd primes diverges, even after finitely many are removed. Otherwise  $\prod_{p \text{ odd}} (1 + 1/p)$  would bound the reciprocal mass of odd squarefree integers; the decomposition  $n = ds^2$  would then make the odd harmonic series converge.

Choose pairwise disjoint finite sets  $P_k$  of odd primes such that  $2^k/4 \leq S_k := \sum_{p \in P_k} 1/p < 2^k/4 + 1$ . Put  $M_k = \prod_{p \in P_k} p$  and  $E = \bigcup_{k \geq 1} \{2^k d : d \mid M_k\}$ . The witness  $\{2\}$  has weighted mass

$$\sum_{k \geq 1} \frac{\prod_{p \in P_k} (1 + 1/p)}{2^{2^k} - 1} \leq 2e \sum_{k \geq 1} \exp(-(\log 2 - 1/4)2^k) < \infty.$$

For the union  $F_m$  of the first  $m$  frames, average  $\log^+ f_{F_m}$  over one period. On  $v_2(n) = k < m$ , the  $k$ th frame contributes  $2^{Z_k(n)}$  divisors, where  $Z_k(n) = \sum_{p \in P_k} \mathbf{1}_{p|n}$ . This event has density  $2^{-(k+1)}$  and is independent of odd-prime divisibility. Thus

$$\mathbb{E}_{F_m} \log^+ f_{F_m} \geq (\log 2) \sum_{k=1}^{m-1} 2^{-(k+1)} S_k \geq \frac{m-1}{8} \log 2.$$

Bound (9) makes these means uniformly bounded for every strengthened cover, so  $E \notin \mathcal{C}$ .

For the reverse direction, choose unused odd primes  $q_j \geq 4^j$  and disjoint finite sets  $P_j$  of unused odd primes. Stop each  $P_j$  at the first prefix with  $R_j := \prod_{p \in P_j} (1 + 1/p) \geq q_j$ . Prime reciprocal divergence permits this, and minimality gives  $q_j \leq R_j < 4q_j/3$ . Put  $S_j = \sum_{p \in P_j} 1/p$ ,  $M_j = \prod_{p \in P_j} p$ , and  $V = \bigcup_{j \geq 1} F_j$ , where  $F_j = \{q_j d : d \mid M_j\}$ . The frames have disjoint prime alphabets, and

$$\sum_{a \in F_j} \frac{1}{a} = \frac{R_j}{q_j} \in [1, 4/3).$$

Thus the reciprocal sum diverges. Every finite prime set misses all factors of all but finitely many frames; each missed frame has base- $b$  weighted mass  $R_j/[q_j(b-1)] \geq 1/(b-1)$ . Hence  $V \notin \mathcal{W}_b$  for every integer  $b \geq 2$ .

For the full infinite-cover cost, from  $\log(1+x) \geq x - x^2/2$  and  $\sum_{p \in P_j} p^{-2} \leq 1$  we have  $S_j \leq \log R_j + 1/2 < \log q_j + 1$ . Set

$$z_j = (j + S_j)^{-1}, \quad \alpha_j = \log_2(1 + z_j), \quad c_{j,q_j d} = z_j^{\omega(d)} \quad (d \mid M_j),$$

with other coefficients zero and  $\omega(d)$  the number of prime factors of  $d$ . If  $q_j \mid n$  and  $Z$  primes in  $P_j$  divide  $n$ , the divisor majorant equals  $(1 + z_j)^Z = f_{F_j}(n)^{\alpha_j}$ ; otherwise both sides vanish. The column cost is  $C_j = q_j^{-1} \prod_{p \in P_j} (1 + z_j/p)$ , whence

$$\frac{C_j 2^{j\alpha_j}}{2^{\alpha_j} - 1} = \frac{(1 + z_j)^j}{q_j z_j} \prod_{p \in P_j} (1 + z_j/p) \leq \frac{e(j + S_j)}{q_j} \leq e((1 + \log 4)j + 1)4^{-j}.$$

The last bound uses that  $(j + \log x + 1)/x$  decreases for  $x \geq 4^j$ . Its sum converges, including the frame-index factor in (V); hence  $V \in \mathcal{C}$ .  $\square$

### 3.6 Combining the two support criteria

Separate small-displacement witnesses need not occur at the same index. For example, a sequence small only at even indices and one small only at odd indices need never have a small sum. The useful feature of (S) is its uniformity in the moving modulus: the positive-cover argument can use the exact observation window selected by the weighted proof.

**Theorem 3.3** (mixed weighted and cover supports). *Let  $E, V \subseteq \mathbb{N}_{>0}$ . Suppose  $E$  satisfies (W) for a finite nonempty prime set  $P$ , and  $V \subseteq \bigcup_j F_j$  for finite sets and nonnegative majorants satisfying the hypotheses of Theorem 3.1, with either (V) or its positive-weight variant. Then  $X_A(b)$  is irrational for every infinite  $A \subseteq E \cup V$  and every integer  $b \geq 2$ .*

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*Proof.* Fix  $\varepsilon > 0$  and an integer  $N_0 \geq 1$ , and put  $\rho = \varepsilon/3$ . Use the cover notation  $B_j, U_j, V_j$  above, with  $\eta_j = 2^{-j}$  in the case (V). Set  $t_j = \rho\eta_j$  and choose  $J$  so that

$$K_J = \sum_{j > J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{16}.$$

Choose a finite  $F \subseteq E$  whose complementary weighted mass is  $\kappa < \rho/16$ . Let  $L$  be a positive common multiple of  $N_0$ , all members of  $F$ , and all members of the first  $J$  finite sets. For large  $H$ , choose the same modulus and averaging length as in the base-two weighted proof:

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \quad G = \lfloor H / \max P \rfloor, \quad M = \lfloor 2^{G/2} \rfloor.$$

The finite estimate (8), with complementary weighted mass  $\kappa$ , remains valid for this  $L$ : its proof requires only that every member of  $F$  divide  $L$ . It also applies to finite or empty  $E$ , since positivity was used only after the averaging estimate. Thus

$$\mathcal{D}_{Q;M,M}(\Delta_{2,E}/\rho) < \frac{1}{8}$$

for sufficiently large  $H$ . For the same finite distribution, (S) and nonnegative interchange give

$$\mathcal{D}_{Q;M,M} S_J \leq (1 + 4Q/M) K_J < \frac{1}{8}, \quad S_J = \sum_{j>J} t_j^{-\alpha_j} V_j,$$

because  $Q/M \rightarrow 0$ . Hence some sample  $N = Qm$  satisfies  $\Delta_{2,E}(N)/\rho + S_J(N) < 1$ . At this index the weighted displacement is below  $\rho$  and  $U_j(N) < t_j$  for every  $j > J$ . The terms with  $j \leq J$  vanish because their exponents divide  $L$ , and hence divide  $N$ . Therefore

$$\Delta_{2,A}(N) \leq \Delta_{2,E}(N) + \Delta_{2,V}(N) < 2\rho < \varepsilon, \quad N \geq Q \geq L \geq N_0.$$

Overlaps between the supports only improve the inequality. Infinitude of  $A$  makes its displacement positive. The atom comparison used in the cover proof transfers arbitrarily small displacements to every integer base; the fixed rational lattice then excludes rationality.  $\square$

The combination is strict as a comparison of sufficient criteria.

**Corollary 3.4** (a host requiring the mixed criterion). *There is an infinite positive support  $U$  with  $U \notin \mathcal{C}$  and  $U \notin \mathcal{W}_b$  for every integer  $b \geq 2$ , such that  $X_A(b)$  is irrational for every infinite  $A \subseteq U$  and every integer  $b \geq 2$ .*

*Proof.* Take  $E, V$  from Proposition 3.2 and put  $U = E \cup V$ . A strengthened cover of  $U$  would cover  $E$ . A finite-prime weighted witness for  $U$  at any fixed base would also witness  $V$ , since its summands are nonnegative. Both conclusions contradict the proposition. Theorem 3.3 applies to the actual pair  $E, V$  and proves the all-base hereditary conclusion; no addition of irrational numbers is involved.  $\square$

The mixed implication has a Lean declaration in the public corpus. Proposition 3.2 and this corollary have ordinary proofs here, but these particular constructions have not been formalised in Lean or independently reviewed. The weighted-only construction follows the first-logarithmic-moment method; the bounded-mass reverse construction was developed in an AI-assisted continuation on 25 September 2026. No historical priority claim is made for the comparison.

Even the fixed-dyadic cover class in (V) is closed under finite unions. For two covers, interleave their  $j$ th frames at positions  $2j - 1, 2j$ , halve each exponent, and retain the nonnegative coefficient columns. The majorants persist since  $f^{\alpha/2} \leq f^\alpha$  for every nonnegative integer  $f$ . Writing  $T_j = C_j 2^{j\alpha_j} / (2^{\alpha_j} - 1)$  for an old column, its new cost is at most

$$\frac{C_j 2^{j\alpha_j}}{2^{\alpha_j/2} - 1} = (1 + 2^{\alpha_j/2}) T_j \leq (1 + \sqrt{2}) T_j.$$

For  $r$  covers, interleave at positions at most  $rj$  and divide each exponent by  $r$ ; the cost ratio is  $(2^\alpha - 1)/(2^{\alpha/r} - 1) < 2r$ . Weighted supports are likewise closed under finite unions: combine their finite prime witnesses, so the prime part grows and  $h/(b^h - 1)$  decreases. Finite supports admit both criteria; subsets inherit each host. Thus the mixed class is closed under finite unions and finite changes using the literal dyadic condition (V), without changing to arbitrary cover weights. Countable unions require a tail budget. Every

prime singleton is admitted, but for the full prime support  $\mathcal{P}$  one has  $\Delta_{2,\mathcal{P}}(N) > 1/3$  for every  $N \geq 1$ : indeed  $\sum_{r \geq 1} 2^{-r} \omega(N+r) \geq 1$ , whereas  $X_{\mathcal{P}}(2) \leq \sum_{a \geq 2} (2^a - 1)^{-1} < 2/3$ . Here  $\omega(n)$  is the number of distinct prime divisors of  $n$ . The reciprocal-summable class is contained in the weighted class, since  $h/(2^h - 1) \leq 1$ . Theorem 1.2 supplies the direct proof of that baseline case.

For comparison, Tao–Teräväinen prove the full-prime case at base 2 [3, Theorem 1.3, p. 4]. The paragraph following it states extensions to prime support at every integer base and to full prime-power support at base 2, leaving the modifications to the reader. It does not treat arbitrary infinite thinnings. The proposed thinning extension is not a premise of any theorem here.

## 4 Finite-support denominator periods

For a finite nonempty  $F \subseteq \mathbb{N}_{>0}$ , let  $D_F$  be the positive reduced denominator of  $X_F(b) = \sum_{n \in F} (b^n - 1)^{-1}$ . We use  $\text{ord}_1(b) = 1$ .

**Theorem 4.1** (the exact denominator period). *Let  $F \subseteq \mathbb{N}_{>0}$  be finite and nonempty, let  $b \geq 2$  be an integer, and let  $D_F > 0$  be the denominator of  $X_F(b)$  in lowest terms. Then  $D_F$  is coprime to  $b$ , and*

$$\text{ord}_{D_F}(b) = \text{lcm}\{n : n \in F\}.$$

*If moreover  $\text{lcm}(F) \geq 2$ , then  $\text{lcm}(F) < D_F$ . We use  $\text{ord}_1(b) = 1$ , so the statement includes  $F = \{1\}$  at  $b = 2$ .*

Put  $L = \text{lcm}(F)$ . Clearing denominators gives  $D_F \mid b^L - 1$ , so the upper divisibility for the order is immediate. For the reverse, choose  $n \geq 2$  maximal under divisibility in  $F$  and a prime  $\ell \mid \Phi_n(b)$ . If  $e = v_\ell(b^n - 1)$ , the full prime power  $\ell^e$  has  $\text{ord}_{\ell^e}(b) = n$ . Every other selected exponent  $m$  has  $n \nmid m$ , hence  $v_\ell(b^m - 1) < e$ . The  $n$ th summand has uniquely smallest  $\ell$ -adic valuation and cannot cancel. Thus  $\ell^e \mid D_F$  and  $n \mid \text{ord}_{D_F}(b)$ . Taking all maximal selected exponents proves the order statement; the size bound follows from  $\text{ord}_{D_F}(b) \mid \varphi(D_F) < D_F$  when  $L \geq 2$ . The case  $F = \{1\}$  has order one directly.

The same unique-valuation argument permits signs  $\pm 1$  on the finite summands. This is an ordinary deduction; it is not formalised. The required cyclotomic prime-power fact is proved in Section 10. The distinction between primes and prime powers is visible in

$$X_{\{2,3\}}(2) = \frac{10}{21}, \quad X_{\{2,6\}}(2) = \frac{22}{63}.$$

In both examples 2 has multiplicative order six modulo the denominator. The first combines orders two and three; the second retains the order-six prime power 9, although no prime divisor of 63 has order six. The boundary  $F = \{1\}$  at base two has  $D_F = L = 1$ . These lower denominator bounds give no upper height control for infinite partial sums and do not decide an infinite-support value. By contrast, Van Assche constructs approximants for the full Lambert series and proves both nonvanishing and decay of the associated integer linear forms [14, Lemma 1 and (34)–(35)]. That approximation mechanism does not follow from denominator survival alone.

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## 5 Rational values and scaled tails

Suppose  $X_A(2) = p/v$ , where  $p \in \mathbb{Z}$  and  $v \geq 1$  is an integer. Multiplying the coefficient-series identity by  $v2^N$  gives

$$z_N := v \sum_{r \geq 1} c_A(N+r)2^{-r} = 2^N p - v \sum_{n=1}^N c_A(n)2^{N-n} \in \mathbb{Z}.$$

Separating the first term of the tail then gives  $z_{N+1} = 2z_N - vc_A(N+1)$ . All the coefficients  $c_A(n)$  are divisor counts of the same set  $A$ .

Two elementary facts delimit what these size estimates can show. Assume first that  $A$  is nonempty and put  $a_0 = \min A$ . Then every  $a_0$  consecutive integers contain a multiple of  $a_0$ , so a divisor-incidence zero window has length at most  $a_0 - 1$ . If  $A$  is infinite, choose  $k$  exponents and a common multiple  $L$ ; then  $c_A(L) \geq k$  and  $\sum_{r \geq 1} c_A(L-1+r)2^{-r} \geq k/2$ , by its first term. Both facts hold without rationality. The arithmetic information lies in the lattice and in the compatibility of all coefficients with the same selector.

Precisely, Dirichlet convolution gives  $\mu * c_A = \mathbf{1}_A$ , where  $\mu$  is the Möbius function. A putative recurrence with integral forcing must therefore satisfy  $(\mu * c_A)(n) \in \{0, 1\}$  for all  $n$ . The companion gives the integer-recurrence criterion, its telescoping proof, Möbius inversion, and the coefficient-series identity in Theorems 6.105–6.107, under *Bounds for a general coefficient sequence*. The example  $A = \{2\}$  there has value  $1/3$ . The correspondence permits finite supports; infinitude remains a separate requirement for a counterexample to Problem 1.3.

## 6 Limits of the small-displacement criterion

The preceding criteria establish irrationality by making  $\Delta_{2,A}(N)$  positive and arbitrarily small. The particular requirement  $\inf_{N \geq 1} \Delta_{2,A}(N) = 0$  fails at full support:

$$\Delta_{2, \mathbb{N}_{>0}}(N) > (2^N - 1) \sum_{a > N} 2^{-a} = 1 - 2^{-N} \geq \frac{1}{2} \quad (N \geq 1).$$

The full-support value is nevertheless [irrational](#) [5]. Thus a proof covering full support cannot rely solely on this small-displacement requirement. This does not exclude other integer linear forms or other uses of averaging.

The distinction also appears in the squarefree support. Its series is irrational at every base  $2^j$ ,  $j \geq 1$ , by Corollary 1.2 and Example 1.1 of Duverney and Tachiya [1, p. 4]. Their corollary covers the  $s$ -free products of any pairwise coprime sequence of polynomial growth, and they present it as support for the conjecture of Erdős and Graham stated here as Problem 1.3. The zero-window obstruction for one normalised certificate scheme concerns that scheme's hypotheses. It gives no obstruction to irrationality of the value. The precise normalisation counterexamples are retained in the companion, together with the complete catalogue of known support families.

A growth-only criterion gives a different class: if  $A = \{c_1 < c_2 < \dots\}$  and  $\limsup_n c_n/2^n = \infty$ , then  $X_A(b)$  is irrational for every integer  $b \geq 2$  [8, Theorem 1]. The companion, Section 1.7, proves the translation from denominator growth and gives

examples showing that this class and reciprocal summability are incomparable. The interval-filling constructions in [9] and the freely chosen lacunary denominators of [16] are not constructions of subseries of these fixed Mersenne weights.

Prime-incidence arguments require additional care as well: multiplying all prime exponents by 2 creates positive correlations between their divisibility indicators. The exact identity and covariance calculation are in the companion, Section 1.5. Size conditions alone do not supply that independence.

## 7 Unique coding and arithmetic membership

Let  $w_n = (2^n - 1)^{-1}$  and

$$\mathcal{A} = \left\{ \sum_{n \geq 1} \varepsilon_n w_n : \varepsilon_n \in \{0, 1\} \right\}, \quad R_N = \sum_{n > N} w_n.$$

The estimate

$$2^{-N} < R_N \leq 2^{-N} + \frac{2}{3} 4^{-N} < w_N \quad (N \geq 1)$$

implies uniqueness of the selector. Indeed, at the first differing exponent  $n$ , the difference  $w_n$  exceeds everything later terms can cancel. The middle inequality follows by writing  $w_n = 2^{-n} + 4^{-n}/(1 - 2^{-n})$  and using  $n \geq N + 1 \geq 2$ . Since every weight exceeds its tail, Hornich's theorem [15], as proved by Nitecki [10, Theorem 4(1), p. 9], shows that the coding image is a Cantor set of measure  $\lim_N 2^N R_N$ ; the estimate above gives  $2^N R_N \rightarrow 1$ , so  $\lambda(\mathcal{A}) = 1$ . Concretely, fixing the first  $N$  digits gives  $2^N$  disjoint closed intervals of length  $R_N$ . These interval unions decrease to  $\mathcal{A}$ , and continuity of Lebesgue measure from above gives the same limit. Kovač–Tao record this strict-tail inequality and the Cantor conclusion in the fixed-base setting [4, Remark 4.1]. For a specified rational target, the useful consequence here is uniqueness of its possible selector. Restricted-set dimension and measure calculations are given in the companion, Section 1.8; Lean proves the [comparison of each restricted tail with the preceding weight](#), the [injectivity of the restricted digit map](#) and the [measure dichotomy](#).

## 8 Greedy membership and an integer recurrence

For  $x \geq 0$ , the greedy rule starts with  $r_0 = x$  and, at rank  $n \geq 1$ , selects  $n$  when  $r_{n-1} \geq (2^n - 1)^{-1}$ , subtracting that weight if selected. Let  $A_x$  be the resulting support and  $c_x(n) = \#\{a \in A_x : a \mid n\}$ . Set

$$P_0 = 0, \quad P_{N+1} = 2P_N + c_x(N+1), \quad Q_N = \lfloor 2^N x \rfloor - P_N.$$

Thus  $P_N = \sum_{j=1}^N 2^{N-j} c_x(j)$  is the integer truncation of the Lambert expansion (2). It is at most  $2^N X_{A_x}(2) \leq 2^N x$ , so  $Q_N \geq 0$ . These integer remainders obey

$$Q_{N+1} = 2Q_N + \beta_N - c_x(N+1), \quad \beta_N = \lfloor 2^{N+1} x \rfloor - 2\lfloor 2^N x \rfloor \in \{0, 1\}. \quad (11)$$

**Theorem 8.1** (membership and nonincreasing integer remainders). *For every real  $x \geq 0$ , the following are equivalent:*

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$$\begin{aligned} x &\in A; \\ \forall K \geq 0 \exists N \geq K : \quad Q_{N+1} &\leq Q_N; \\ \forall K \geq 0 \exists N \in [K, K + 2\lfloor \sqrt{K} \rfloor + 12) : \quad Q_{N+1} &\leq Q_N. \end{aligned}$$

*Proof.* Strict domination of each Mersenne weight over its tail implies that a represented target is recovered by the greedy rule. For such a target,

$$0 \leq Q_N \leq \sum_{r \geq 1} c_x(N+r)2^{-r} \leq 2\sqrt{N} + 4,$$

since  $c_x(n) \leq \tau(n) \leq 2\sqrt{n}$  and  $\sqrt{N+r} \leq \sqrt{N} + \sqrt{r} \leq \sqrt{N} + r$ . Put  $s = \lfloor \sqrt{K} \rfloor$  and  $T = 2s + 12$ . Strict increase at all  $T$  steps would imply  $Q_{K+T} \geq T$ . But  $K + T < (s + 4)^2$  gives  $Q_{K+T} < 2s + 12 = T$ . This proves the window condition and hence cofinal nonincreases.

Conversely, if  $x \notin A$ , then  $\delta = x - X_{A_x}(2) > 0$ . The Lambert prefix is at most  $2^N X_{A_x}(2)$ , so  $Q_N \geq 2^N \delta - 1$ . Eventually this exceeds  $c_x(N+1) \leq N+1$ . Equation (11) then makes  $Q_N$  strictly increasing, contradicting cofinal nonincreases.  $\square$

Theorem 8.1 characterises membership, but it does not locate the required nonincreases for a specified rational target. For  $x = 0$ , the selector is empty and  $Q_N = 0$  at every rank. A finite subseries sum is likewise represented and satisfies the criterion. In contrast,  $x = 3/4$  lies strictly between  $R_1 < 2/3$  and  $w_1 = 1$ , so it is not represented and its integer remainders eventually increase strictly. For  $1/2$  and  $1/21$ , the theorem does not establish which alternative occurs.

The square-root estimate is uniform in the target. There is also a uniform subpower refinement: for each  $0 < \varepsilon < 1$ , a constant  $C_\varepsilon$  gives the window length  $\lceil C_\varepsilon(K+1)^\varepsilon \rceil$ . Indeed,  $\tau(n) \leq A_\varepsilon n^\varepsilon$  gives  $Q_N \leq D_\varepsilon(N+1)^\varepsilon$  for represented targets, where  $D_\varepsilon = A_\varepsilon \sum_{r \geq 1} r^\varepsilon 2^{-r}$ . Choose  $C_\varepsilon > D_\varepsilon(C_\varepsilon + 3)^\varepsilon$ , which is possible because  $\varepsilon < 1$ . For  $T = \lceil C_\varepsilon(K+1)^\varepsilon \rceil$ , strict increase at all  $T$  steps would give

$$T \leq Q_{K+T} \leq D_\varepsilon(K+T+1)^\varepsilon \leq D_\varepsilon(C_\varepsilon + 3)^\varepsilon (K+1)^\varepsilon < T,$$

a contradiction. The converse still follows from exponential growth when  $x$  is not represented. This ordinary argument strengthens the window bound, not the occurrence claim for a specified target; only the square-root window of Theorem 8.1 has a Lean proof.

At  $x = 1/2$ , the digits  $\beta_N$  vanish for  $N \geq 1$ ; at  $x = 1/21$  they are six-periodic. Thus the unresolved arithmetic input is

$$\boxed{\forall K \exists N \geq K : \quad c_x(N+1) \geq Q_N + \beta_N, \quad x \in \{1/2, 1/21\}.} \quad (12)$$

Both  $Q_N$  and  $c_x$  must arise from the same greedy selector. The companion retains exact counterexamples to fixed-multiplier repair schedules and the finite phase masks used to test them. Neither target is decided here.

## 9 Further questions

The support theorems give sufficient conditions for irrationality; the tests below instead concern whether a specified rational is a subseries sum. Their hypotheses must be verified for the actual selector or for finite approximants at arbitrarily large depths. An equivalent test does not itself establish membership.

**Integer quotients for  $1/21$ .** For  $M, d \geq 1$ , set

$$q_M(d) = \left\lfloor \frac{2^M}{2^d - 1} \right\rfloor, \quad T_M = \left\lfloor \frac{2^M}{21} \right\rfloor.$$

Starting with the integer remainder  $T_{2R}$ , consider  $d = 2, \dots, R$  in order and subtract  $q_{2R}(d)$  whenever it does not exceed the current remainder. Let  $D_R$  be the set of selected exponents and  $s_R$  the final remainder. Thus

$$s_R = T_{2R} - \sum_{d \in D_R} q_{2R}(d) \geq 0.$$

A second comparison uses the same integer rule through  $d = 2R$ , rather than stopping at  $R$ . For example,  $R = 6$  gives  $T_{12} = 195$ ,  $D_6 = \{5\}$  and  $s_6 = 195 - \lfloor 4096/31 \rfloor = 63 \leq 64$ . This is one row satisfying the bound below, not evidence that such rows occur at unbounded depths.

Write  $r_n = r_n(1/21)$  for the real greedy remainder and  $A_{1/21}$  for its support. The condition  $\mathcal{F}_{21}$  means that there exist integers  $n, R_0, K_0 \geq 0$  with all of the following properties:

1.  $r_n > R_n$ , the set of positive exponents outside  $A_{1/21}$  is finite, and every exponent greater than  $n$  belongs to  $A_{1/21}$ ;
2. for every  $R \geq R_0$ , the integer rule with target  $T_{2R}$  and weights  $q_{2R}(2), \dots, q_{2R}(2R)$  selects exactly the same exponents as the real greedy rule on  $\{2, \dots, 2R\}$ ;
3. for every  $K \geq K_0$ , the interval  $(K, 2K]$  contains an exponent of  $A_{1/21}$ .

Some of these clauses follow from others once the eventual all-selected behaviour is known. They are written out to match the precise branch used in the formal result. In particular,  $\mathcal{F}_{21}$  is not a hypothesis about an arbitrary recurrence with similar coefficients.

**Theorem 9.1** (integer-quotient tests for  $1/21$ ). *The following statements hold.*

1.  $1/21 \in \mathcal{A}$  if and only if  $\mathcal{F}_{21}$  does not hold.
2. If there is an unbounded sequence of ranks  $R$  with  $s_R \leq 2^R$ , then  $1/21 \in \mathcal{A}$ .
3. On  $\mathcal{F}_{21}$ , eventually  $s_R > 2^R$ , the boundary rank  $R + 1$  belongs to  $D_{R+1}$ , and, for all sufficiently large  $R$ ,

$$\begin{aligned} D_{R+1} &= D_R \cup \{R + 1\}, \\ s_{R+1} &= 4s_R + \left\lfloor \frac{4(2^{2R} \bmod 21)}{21} \right\rfloor - 2c_{D_R}(2R + 1) - c_{D_R}(2R + 2) \\ &\quad - (2^{R+1} + 1). \end{aligned}$$

Here  $c_{D_R}(m) = \#\{d \in D_R : d \mid m\}$ .

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The finite uniqueness statement is also explicit: if  $D \subseteq \{2, \dots, R\}$  and an integer  $s$  satisfy  $\sum_{d \in D} q_{2R}(d) + s = T_{2R}$  with  $0 \leq s \leq 2^R$ , then  $D = D_R$  and  $s = s_R$ . This is the denominator-specific separation theorem ([uniqueness of a finite representation with the stated remainder bound](#)). The cited finite crossing lemmas give additional consequences under their alignment hypotheses: an earlier finite prefix cannot occur, and a real greedy exponent must be skipped ([the missing-prefix consequence of an aligned crossing](#), [the real greedy skip forced by that crossing](#)).

**Approximation to 1/2 by finite supports.** For  $A \subseteq \{2, \dots, M\}$  and  $1 \leq m \leq M$ , define the integer

$$K_A(m) = 2^{m-1} - \sum_{j=2}^m 2^{m-j} c_A(j).$$

It measures the error in the truncated divisor-coefficient sum, after multiplication by  $2^m$ . In particular,  $K_A(1) = 1$  and  $K_A(m+1) = 2K_A(m) - c_A(m+1)$ .

**Theorem 9.2** (finite approximations with vanishing scaled error). *Suppose there are integers  $M_j \geq 1$  tending to infinity and sets  $A_j \subseteq \{2, \dots, M_j\}$  such that*

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$$\frac{|K_{A_j}(M_j)|}{2^{M_j}} \rightarrow 0.$$

*Then  $X_A(2) = 1/2$  for some infinite set  $A \subseteq \mathbb{N}_{>0}$ .*

No agreement between different  $A_j$  is assumed, and no bound is imposed on their earlier carries. Producing such finite approximants would refute Problem 1.3; their existence is not established here. The implication follows from the estimate

$$\left| X_{A_j}(2) - \frac{1}{2} \right| \leq \frac{|K_{A_j}(M_j)| + 2\sqrt{M_j} + 4}{2^{M_j}}.$$

It follows by separating the Lambert-series coefficient tail and using  $c_{A_j}(n) \leq \tau(n) \leq 2\sqrt{n}$ . The finite sums therefore tend to 1/2. The achievement set is closed, and no finite support has value 1/2, so its representing support is infinite. A square-root bound on the terminal carry would suffice, but the theorem permits any error  $o(2^{M_j})$ . For a finite  $D \subseteq \{2, \dots, M\}$ , the floor-quotient identity gives

$$K_D(M) = 2^{M-1} - \sum_{d \in D} \left\lfloor \frac{2^M}{2^d - 1} \right\rfloor.$$

Thus an exact quotient row has  $K_D(M) = 1$ , whereas the terminal criterion permits a larger error of either sign. These are different conditions at a fixed depth, even though their cofinal existence conditions both characterise half-membership.

Conversely, if  $X_A(2) = 1/2$ , then  $1 \notin A$ , and the prefixes  $A_M = A \cap \{2, \dots, M\}$  satisfy

$$K_{A_M}(M) = \sum_{r \geq 1} c_A(M+r) 2^{-r} \leq 2\sqrt{M} + 4.$$

The equality uses the coefficients of the full support  $A$  on the right: truncation does not change the coefficients through  $M$ . Hence the existence hypothesis of Theorem 9.2 is equivalent to half-membership. Allowing incompatible finite supports removes a construction requirement, not the difficulty of the existence problem. Compactness proves the implication; it does not produce the approximating supports.

**Finite families with a shared prefix.** Put  $B(m) = 2\lfloor \sqrt{m} \rfloor + 4$ . For  $0 \leq K \leq M$ , consider a family  $A_1, \dots, A_{B(M)} \subseteq \{2, \dots, M\}$  with

$$1 \leq K_{A_k}(m) \leq B(m) \quad (1 \leq m \leq M), \quad K_{A_k}(M) = k.$$

Require that all the  $A_k$  agree on  $\{1, \dots, K\}$  and that some integer  $E \geq B(M)$  satisfies

$$\sum_{j=K+1}^M 1_{A_k}(j) 2^{M-j} + k = E \quad (1 \leq k \leq B(M)).$$

Thus the binary suffixes have consecutive values  $E - k$ , while the terminal carry ranges through the entire permitted interval. This is substantially more data than one finite approximation to  $1/2$ .

**Theorem 9.3** (unbounded shared-prefix families imply a half-support). *Suppose that for every  $N$  there are  $M, K$  with  $\max\{N, 1\} \leq M$ ,  $0 \leq K \leq M$ , and a family satisfying all the conditions in the preceding paragraph. Then  $X_A(2) = 1/2$  for some infinite set  $A \subseteq \mathbb{N}_{>0}$ .*

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Choosing one member from each family gives the terminal bound in Theorem 9.2, proving the conclusion. The shared-prefix conditions are needed for the proposed construction of the families, not for this final compactness step. Such a construction at unbounded depths would refute Problem 1.3; none is proved here.

**Problem 9.4** (membership of  $1/21$  in the Mersenne achievement set). For the greedy remainder of  $1/21$ , prove that arbitrarily large  $N$  satisfy

$$2^{N+1} r_N(1/21) < \frac{2^{N+1}}{2^{N+1} - 1}, \quad \text{equivalently} \quad r_N(1/21) < \frac{1}{2^{N+1} - 1}.$$

Equivalently, exclude the condition  $\mathcal{F}_{21}$  defined above. One sufficient route is to rule out the eventual recurrence in Theorem 9.1(3) together with  $s_R > 2^R$ ; another is to prove  $s_R \leq 2^R$  at arbitrarily large ranks. Neither sufficient route is claimed to be necessary by itself.

**Problem 9.5** (bounded returns of the scaled remainder). For each of the targets  $x = 1/2$  and  $x = 1/21$ , does the greedy remainder  $r_N(x)$  return to one bounded interval after scaling by  $2^N$ ?

$$\exists B < \infty \forall K \exists N \geq K : \quad 2^N r_N(x) \leq B.$$

**Problem 9.6** (arithmetic tests for the greedy sequence). Can a finite-memory, 2-adic or discrepancy argument prove  $s_R \leq 2^R$  at arbitrarily large ranks, or rule out the eventual recurrence in Theorem 9.1(3)? Such an argument must use the divisor counts of the actual greedy support. Can a bounded window of  $R \bmod 6$ , residues of  $s_R$ , endpoint

divisor counts and the finite set of eventual skips force a decrease or a contradiction? Alternatively, can one show that these bounded-memory data cannot distinguish the actual sequence from sequences that remain above  $2^R$  but need not come from a support?

**Problem 9.7** (final-skip Diophantine exclusion). Let  $E = \sum_{n \geq 1} (2^n - 1)^{-1}$ . If  $1/21 \notin A$ , let  $M$  be the last skipped exponent,  $S_M$  its finite skipped prefix, and

$$a_M = \frac{1}{21} + \sum_{d \in S_M} \frac{1}{2^d - 1}.$$

Write  $\text{gap}_M = (2^M - 1)^{-1} - R_M > 0$ . Can the arithmetic restrictions on this last skipped exponent be used to prove  $|E - a_M| \geq \text{gap}_M$ , contradicting

$$0 < a_M - E < \text{gap}_M?$$

## 10 Elementary details used in the proofs

**Cyclotomic prime-power fact.** Let  $b \geq 2$ ,  $n \geq 2$ ,  $\ell \mid \Phi_n(b)$  be prime and  $e = v_\ell(b^n - 1)$ . Then  $\text{ord}_{\ell^e}(b) = n$ . Here is a proof including the exceptional 2-adic behaviour. For odd  $\ell$ , put  $d = \text{ord}_\ell(b)$  and  $s = v_\ell(b^d - 1)$ . The elementary lifting identity  $v_\ell(b^{d^t} - 1) = s + v_\ell(t)$  follows by factoring a geometric sum when  $\ell \nmid t$  and by a binomial expansion for a factor  $\ell$ . In the factorisation  $b^n - 1 = \prod_{r \mid n} \Phi_r(b)$ , subtracting these valuations over proper divisors gives

$$v_\ell(\Phi_n(b)) = \begin{cases} s & n = d, \\ 1 & n = d\ell^j, j \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

If  $n = d$ , no smaller positive exponent gives  $b^m \equiv 1 \pmod{\ell}$ . If  $n = d\ell^j$ , the full valuation is  $e = s + j$ ; the same lifting identity shows that the least exponent giving valuation  $e$  is  $d\ell^j = n$ . For  $\ell = 2$ ,  $b$  is odd. Factoring  $b^{2^t} - 1$  gives

$$v_2(b^{2^j} - 1) = v_2(b - 1) + v_2(b + 1) + j - 1 \quad (j \geq 1).$$

The only possible indices  $n \geq 2$  are  $n = 2^j$ :  $v_2(\Phi_2(b)) = v_2(b + 1)$  and  $v_2(\Phi_{2^j}(b)) = 1$  for  $j \geq 2$ ; all other indices have valuation zero. The displayed identity again makes  $2^j$  the least exponent with the full valuation. This proves the fact. Finally  $\Phi_n(b) > 1$ : each primitive  $n$ th root  $\zeta$  satisfies  $|b - \zeta| > 1$ , and their product is  $\Phi_n(b)$ .

**Exact scalar cover cost.** For  $t \geq 1$ , put  $z = \log_2 t$ . The substitution  $y = 2^\alpha$  reduces the infimum defining  $\Psi(t)$  to  $y^z/(y - 1)$  on  $1 < y \leq 2$ . Its derivative has the sign of  $(z - 1)y - z$ . Hence

$$\Psi(t) = \begin{cases} t, & 1 \leq t \leq 4, \\ z^z/(z - 1)^{z-1}, & t > 4. \end{cases}$$

The endpoint and the open lower bound  $\alpha > 0$  are both included in this calculation. This scalar identity does not supply a converse to the positive-cover criterion.

## A Sources

Erdős’s five-page paper was checked directly: p. 222 states the reciprocal-summable extension without proof, and p. 226 describes the fractional-part approach. The original Luca–Tachiya and Hornich articles were not independently retrieved. The periodic theorem [12] was checked in Luca and Tachiya’s own account [11, Theorem A and Example 2, pp. 139–140], and the strict-tail result in Nitecki’s exposition. The *Formal Conjectures* file [6] is statement-level prior art, not a proof dependency.

The evidence record lists the Lean declarations behind each marked result, with their versions and independent checks. The Lean sources for the supplementary statements quoted from the companion (finite sums and denominators, the achievement set and its restrictions, squarefree divisor counts, half-membership and the tests for  $1/21$ ) are linked beside the corresponding statements there.

**Reproducing the weighted theorem.** The two Lean declarations for Theorem 1.1 are both in the public source at [commit 91ca3405](#). That snapshot pins Lean 4.29.1 in `lean-toolchain` and its Mathlib revision in `lake-manifest.json`. From a complete checkout, with `elan` installed, build the two modules with:

```
lake exe cache get
python3 scripts/lean_fast_build.py --jobs 2 \
  ErdosProblems.Erdos257.PaperCompleteR8.WeightedReturn \
  ErdosProblems.Erdos257.PaperCompleteR8.WeightedHereditaryClaim
```

The cache download is optional. The modules contain `divisibilityWeightedClaim` and `finitePrimeWeighted_fixedBase_hereditary`, respectively; the evidence record linked above gives their exact statements and the separate Comparator checks.

**Library declarations.** Several statements of this note and its companion rest on the following library declarations, each opened at a fixed revision. Pairwise-coprime supports: [factorial-support instance](#); [powers-of-two-support instance](#); [pairwise-coprime theorem](#). Eventually periodic weights: [irrationality for the stated eventually periodic weights](#). Fixed-core supports: [carry certificates for the stated fixed-core support](#); [irrationality for the stated fixed-core support](#). Finite supports and denominators: [noncollapse for the reduced rational denominator](#). Rational values and carries: [rational support values and Möbius-inverted carries](#); [rationality and integer recurrences with vanishing scaled error](#); [a lower bound for reciprocal mass](#); [unboundedness of the shifted odd-tail recurrence](#); [lengths of intervals with zero divisor count](#). Composite dilation: [the divisor-count identity under dilation](#); [the upper bound for the dilation correction](#); [the example with exponents two and six](#). The achievement set: [Lebesgue measure one](#); [perfect](#); [totally disconnected](#); [nowhere dense](#); [membership and greedy tail bounds](#); [non-membership of  \$3/4\$](#) ; [the greedy form](#); [the finite-support exclusion](#). Restricted achievement sets: [the formal dichotomy](#). Greedy skips and exact rows: [the cofinal-skip hypothesis](#); [an exact finite sum from one greedy skip](#); [exact finite sums from arbitrarily late greedy skips](#); [membership from infinitely many greedy skips](#); [the upper-half Boolean fill](#). Membership of  $1/2$ : [the terminal-bit form](#); [the skipped-rank form](#). The two-thirds band: [general band localization](#); [two-thirds band](#); [integral safety](#); [odd numerator bound](#).

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